[Name: Shantanu Singh Rao / **Roll No.:** M24DE2026**]**

Project Title

Scalable Analysis and Predictive Modeling on Large-Scale E-commerce

1. BROAD OVERVIEW

This project centers on large-scale data processing, exploratory analysis, and predictive modeling applied to a high-volume e-commerce transaction dataset, leveraging Python-based Big Data tools. It aims to replicate real-world industry conditions, where massive datasets—comprising millions of records—require efficient cleaning, transformation, analysis, and modeling. The work will showcase scalable data ingestion, distributed processing, feature engineering, and machine learning model development through frameworks like PySpark and Pandas. Key outcomes include insights on customer behavior, sales forecasting, and purchase prediction, all while managing large datasets effectively.

2. OBJECTIVES

• Efficiently process and analyze large-scale datasets with Python-based Big Data tools.

• Conduct data cleaning, transformation, and feature engineering across millions of records.

• Develop and assess predictive models for sales forecasting or customer purchase prediction.

• Evaluate performance differences between traditional processing (Pandas) and distributed processing (PySpark).

3. TECH STACK AND JUSTIFICATION

Python serves as the core programming language, thanks to its robust ecosystem for data analysis and machine learning. PySpark enables distributed data processing to manage large datasets beyond the limits of standard in-memory tools. Pandas and NumPy handle data manipulation and numerical operations, while Matplotlib and Seaborn create visualizations of large-scale insights. Scikit-learn powers machine learning model implementation, with Apache Spark MLlib as an optional choice for scalable training. This stack delivers both scalability and in-depth analytics.

4. DELIVERABLES (FINAL EVALUATION CRITERIA)

• Cleaned and preprocessed large-scale dataset.

• Exploratory Data Analysis (EDA) report featuring visualizations.

• PySpark-based implementation for scalable data processing.

• Predictive machine learning model, complete with performance metrics.

• Report comparing processing times and efficiency.

• Final project report (PDF/Word format) and source code (Jupyter Notebook).

5. DATA SOURCE, FORMAT, AND SIZE

The dataset comes from public sources like Kaggle's E-commerce transactions or retail sales datasets. It will be in CSV format, encompassing millions of transaction records with attributes such as transaction ID, customer ID, product category, purchase amount, timestamp, payment method, and geographic location. Expected size ranges from 500MB to 2GB to mimic real Big Data environments. Synthetic data can be generated if needed to expand volume for scalability tests.